
A Median Perspective on Unlabeled Data for Out-of-Distribution Detection

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Reliable deployment requires models to recognize when an input falls outside
2 the distribution represented by their labeled training data. Unlabeled in-the-wild
3 data offers a practical source of auxiliary evidence for this problem, but it is also
4 statistically ambiguous: the available pool is typically a contaminated mixture of
5 in-distribution (InD) and out-of-distribution (OOD) samples rather than a labeled
6 or clean OOD corpus. We present Medix, a median-centric framework that turns
7 such mixtures into candidate outlier evidence and uses the extracted candidates,
8 together with labeled InD data, to train an OOD detector. Medix exploits the
9 robustness of element-wise medians in gradient space: samples whose removal
10 brings the median wild gradient closer to an InD reference are treated as likely OOD
11 candidates. We provide error bounds for the filtering stage that make explicit how
12 contamination, concentration, and gradient separation determine inlier removal
13 and OOD retention under the stated assumptions. Across the evaluated CIFAR-
14 based open-world benchmarks, Medix improves OOD detection over the compared
15 baselines, supporting median-based filtering as a simple and principled mechanism
16 for leveraging unlabeled wild data.

17 1 Introduction

18 Machine learning systems deployed in open environments face reliability requirements that closed-set
19 evaluation does not adequately capture. A particularly consequential failure mode is the treatment of
20 out-of-distribution (OOD) inputs as ordinary in-distribution (InD) examples: samples drawn from
21 categories outside the training label space should trigger abstention or rejection rather than forced
22 class assignment. Nevertheless, neural networks are frequently overconfident on such inputs and
23 generally lack an intrinsic mechanism for recognizing them at inference time (Nguyen et al., 2015).

24 OOD detection is therefore a necessary component of open-world prediction, yet it is difficult precisely
25 because the relevant unknown distributions are absent from conventional supervised training. Recent
26 work (Katz-Samuels et al., 2022a; Du et al., 2024a) narrows this gap by incorporating additional
27 “in-the-wild” data into training. In particular, Katz-Samuels et al. (2022a) showed that unlabeled
28 wild data can regularize learning while preserving the primary objective of labeled InD classification.
29 Such data is attractive because it can be collected from the operating environment after deployment,
30 exposing the learner to variation that is closer to the OOD inputs encountered at test time than a
31 purely closed training set.

32 The practical value of wild data comes with a central statistical ambiguity. The unlabeled pool is
33 a mixture of InD and OOD samples, and the learner does not observe which component produced
34 any individual point. Methods that require a clean auxiliary OOD dataset are therefore mismatched
35 to this setting, while methods that use wild data must extract outlier evidence without relying on
36 OOD labels. Despite its practical importance, this problem remains comparatively underdeveloped,

particularly on the theoretical side; to the best of our knowledge, Du et al. (2024a) is the only prior work that provides a formal foundation for the “in-the-wild” setting.

Median-based procedures offer a natural statistical tool for this contaminated-mixture regime. Recent data-pruning results show that median estimates can remain stable in the presence of anomalous samples (Acharya et al., 2024). This robustness suggests that, rather than treating wild data as an undifferentiated auxiliary set, one can identify samples whose gradient behavior disrupts the central InD tendency. Motivated by this connection, we ask:

How can median-based methods leverage unlabeled wild data to facilitate OOD detection with theoretical guarantees?

We answer this question with Medix, a median-centric formulation of OOD detection from unlabeled wild data. Medix first defines an InD reference gradient from labeled data, then greedily removes wild samples whose exclusion most improves the alignment between this reference and the element-wise median of wild-sample gradients. The removed samples become candidate OOD examples for detector training. Our analysis studies the median filtering objective and gives high-probability error bounds for the filtering stage, making explicit the assumptions under which robust extraction is possible. In particular, the inlier error decomposes into a contamination effect, which remains controlled when the OOD proportion is below 50%, and a concentration effect, which limits deviations of InD gradients under sub-Gaussian behavior. These terms clarify why the median criterion can remain stable in contaminated mixtures and when InD samples are unlikely to be incorrectly removed.

We evaluate Medix against two families of baselines: methods trained only on InD data and methods trained with both InD data and auxiliary unlabeled data. On CIFAR-100 (Krizhevsky et al., 2009), Medix improves over the strong KNN+ baseline (Sun et al., 2022) by an average of 40.98% in FPR95. Unlike Outlier Exposure (Hendrycks et al., 2019), which relies on an auxiliary unlabeled set that is assumed to be cleanly separable from the InD data, Medix operates directly on mixed wild data. Relative to WOODS (Katz-Samuels et al., 2022a), Medix reduces average FPR95 by 1.32% on CIFAR-100 and 2.60% on CIFAR-10, showing that the median-based filtering stage translates into measurable detection gains under the same broad wild-data protocol.

The paper makes three contributions.

- C1)** We introduce Medix, a median-centric greedy framework whose central technical component is a gradient-space filtering stage that extracts candidate outliers from unlabeled wild data before training an OOD detector with those candidates and labeled InD samples.
- C2)** We establish theoretical guarantees for median-based filtering in unlabeled mixtures, bounding both InD samples incorrectly flagged as outliers and OOD samples incorrectly retained as inliers. The bounds separate the contamination, concentration, and separation effects that govern robustness, including the regime in which the OOD proportion is below 50%.
- C3)** We evaluate Medix across eleven InD-OOD pairs against 20 competitive baselines. The results show consistent improvements in the reported comparisons and demonstrate that the outlier extraction step can achieve a low error rate (e.g. only 12.5%; see Figure 2), empirically supporting the role of median filtering in the overall detector.

2 Preliminaries and Problem Setup

This section defines the OOD detection problem studied in the paper. We specify the labeled InD sample, the unlabeled wild mixture, the classifier and detector parameterizations, and the metrics used to evaluate both rejection performance and InD predictive accuracy.

Labeled In-Distribution Data. Let $\mathcal{X}$ denote the input space and let $\mathcal{Y} = \{1, \dots, K\}$ denote the set of InD labels. A labeled dataset $\mathcal{S}_{\text{in}} = \{(x_1, y_1), \dots, (x_n, y_n)\}$ contains n independently and identically distributed (i.i.d.) samples from $\mathbb{P}_{XY}$, an unknown joint distribution on $\mathcal{X} \times \mathcal{Y}$. We denote by $\mathbb{P}_{\text{in}}$ the marginal distribution of $\mathbb{P}_{XY}$ over $\mathcal{X}$; this marginal defines the InD input distribution used to train the InD model from $\mathcal{S}_{\text{in}}$.

Out-of-distribution detection. We consider a deployment regime in which a model trained on labeled InD data may encounter inputs whose semantic labels are outside the training label space, i.e., for some label $y \notin \mathcal{Y}$. For these inputs, the desired behavior is abstention or rejection rather

than forced assignment to an InD class. Thus, while the classifier remains multi-class over $\mathcal{Y}$, the detection objective is binary: decide whether an input is drawn from $\mathbb{P}_{\text{in}}$ or from an OOD source.

Unlabeled wild data. A central obstacle in OOD detection is that labeled OOD samples are difficult to enumerate, collect, and annotate because the OOD sample space can be extremely broad. Following Katz-Samuels et al. (2022a), we treat unlabeled wild data, $\mathcal{S}_{\text{wild}} = \{\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_m\}$, as the available auxiliary source after a model trained on $\mathcal{S}_{\text{in}}$ is exposed to its operating environment. This set is not a clean OOD corpus; it is an unlabeled mixture of InD and OOD inputs. Consistent with Du et al. (2024a); Katz-Samuels et al. (2022a), we represent its marginal distribution using the Huber contamination model (Huber, 1964)

$$\mathbb{P}_{\text{wild}} := (1 - \pi)\mathbb{P}_{\text{in}} + \pi\mathbb{P}_{\text{out}}, \quad \pi \in (0, 1], \quad (1)$$

where π denotes the OOD contamination proportion and $\mathbb{P}_{\text{out}}$ is the OOD distribution over $\mathcal{X}$. The case $\pi = 0$ contains no OOD samples and therefore lies outside the nontrivial detection regime considered here.

Models and Loss Functions. Let $f_\phi : \mathcal{X} \rightarrow \mathbb{R}^K$ be the InD classifier with parameters $\phi \in \Phi$, where Φ is the classifier parameter space. Its output represents a soft probability distribution over the $K = |\mathcal{Y}|$ InD classes, and the supervised InD loss is $\ell : \mathbb{R}^K \times \mathcal{Y} \rightarrow \mathbb{R}$. For detection, we use a separate binary detector $g_\theta : \mathcal{X} \rightarrow \mathbb{R}$ with parameters $\theta \in \Theta$. The binary loss is written $\ell_b(g_\theta(x), y_b)$ for labels $y_b \in \mathcal{Y}_b := \{y_+, y_-\}$, where $y_+ > 0$ denotes the InD decision and $y_- < 0$ denotes the OOD decision.

Learning objective. The learning objective combines the supervised InD signal from $\mathbb{P}_{\text{in}}$ with the unlabeled mixture information in $\mathbb{P}_{\text{wild}}$ to train the OOD detector g_θ while retaining the multi-class classifier f_ϕ . At test time, detection and classification are evaluated by the following metrics:

$$\begin{aligned} \downarrow \text{FPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{out}}^{\text{test}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{TPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{in}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{Acc}(f_\phi) &= \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{\mathcal{X}\mathcal{Y}}} [\mathbb{I}\{f_\phi(\mathbf{x}) = y\}], \end{aligned}$$

where $\mathbb{I}\{\cdot\}$ is the indicator function and $\mathbb{P}_{\text{out}}^{\text{test}}$ denotes the OOD distribution used at evaluation time.

3 Method: Median-Centric Framework for OOD Detection

This section presents **Medix**, a median-centric framework for OOD detection from unlabeled wild data. Instead of relying solely on labeled InD data or assuming a clean auxiliary OOD set, Medix converts a contaminated wild mixture into candidate outlier evidence for detector training. The framework has two stages: **1) Outlier Extraction**: a median-based filtering procedure that isolates candidate OOD samples from the unlabeled wild data (Section 3.1); and **2) Detector Training**: binary OOD detector training using labeled InD data together with the candidates extracted in the first stage (Section 3.2). For the second stage, we follow the protocol introduced by Du et al. (2024a). This organization highlights the main technical contribution of Medix: the filtering stage supplies usable negative evidence without requiring labels for the wild samples.

3.1 Extracting Candidate Outliers from the Wild Data

To extract candidate outliers from the wild mixture $\mathcal{S}_{\text{wild}}$, Medix defines a filtering criterion on gradients of a classifier trained only on the labeled InD dataset $\mathcal{S}_{\text{in}}$. The gradient representation makes it possible to compare wild samples with an InD reference while avoiding any dependence on labeled OOD examples.

Reference gradient estimation from InD data. Medix first estimates the InD reference gradient from $\mathcal{S}_{\text{in}}$. The classifier f_ϕ is trained on labeled InD data by empirical risk minimization (ERM):

$$\phi_{\mathcal{S}_{\text{in}}} \in \arg \min_{\phi \in \Phi} \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi), \quad \text{where } \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi) = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \ell(f_\phi(\mathbf{x}_i), y_i), \quad (2)$$

where $\phi_{\mathcal{S}_{\text{in}}}$ denotes the learned parameters. After training, we compute the mean InD gradient $\bar{\nabla}_{\text{in}}$ by averaging the loss gradients with respect to the model parameters over the labeled InD sample:

$$\bar{\nabla}_{\text{in}} = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}}(\mathbf{x}_i), y_i). \quad (3)$$

The vector $\bar{\nabla}_{\text{in}}$ therefore serves as the reference statistic against which Medix evaluates the gradient behavior of wild samples.

Motivation. The filtering criterion rests on the empirical observation that adding OOD samples to a wild set displaces an aggregate wild-gradient statistic away from the InD reference. We examine this behavior using CIFAR-10 (Krizhevsky et al., 2009) as the InD dataset and SVHN (Netzer et al., 2011) as the OOD dataset. The initial $\mathcal{S}_{\text{wild}}$ contains 10,000 CIFAR-10 samples disjoint from the training set used to learn $\phi_{\mathcal{S}_{\text{in}}}$ and compute $\bar{\nabla}_{\text{in}}$. We then incrementally add SVHN OOD samples to $\mathcal{S}_{\text{wild}}$ and measure the L_2 deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median (EWM) of the wild-sample gradients:

$$\left\| \bar{\nabla}_{\text{in}} - \text{EWM} \left(\left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}_{\text{wild}}} \right) \right\|.$$

Figure 1 shows a clear monotonic increase in this deviation as OOD samples are added, supporting median-gradient displacement as an outlier signal in this setting. The same pattern motivates the stopping rule in Medix: during filtering, the procedure terminates once the reduction in L_2 deviation between iterations falls below the threshold ϵ .

Filtering potential outliers from unlabeled wild data. Motivated by Figure 1, we formulate outlier extraction by identifying the subset of wild samples whose median gradient is closest to the InD reference:

$$\mathcal{S}_{\text{in}}^* = \arg \min_{\mathcal{S} \subseteq \mathcal{S}_{\text{wild}}} \left\| \bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}}) \right\|, \text{ where } G_{\mathcal{S}} = \left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}}. \quad (4)$$

Here $\text{EWM}(\cdot)$ denotes the element-wise median and $\hat{y}_{\tilde{\mathbf{x}}_i}$ is the predicted label for wild sample $\tilde{\mathbf{x}}_i$. For notational simplicity, we write $G_{\mathcal{S}} = \{ \nabla \ell(\tilde{\mathbf{x}}_i) \}_{i \in \mathcal{S}}$. The objective selects a subset $\mathcal{S}$ of $\mathcal{S}_{\text{wild}}$ whose EWM gradient best matches $\bar{\nabla}_{\text{in}}$. In light of Figure 1, the minimizer $\mathcal{S}_{\text{in}}^*$ is interpreted as the putative InD component of the wild mixture, while its complement $\mathcal{S}_{\text{out}}^* = \mathcal{S}_{\text{wild}} \setminus \mathcal{S}_{\text{in}}^*$ provides the corresponding candidate OOD set.

Solving (4) exactly can be computationally prohibitive as $\mathcal{S}_{\text{wild}}$ grows. Medix therefore uses the greedy leave-one-out approximation summarized in Algorithm 1: it iteratively removes samples whose exclusion most improves alignment between the wild-set median gradient and the InD reference.

At iteration t , the algorithm computes $d_t = \left\| \text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}} \right\|$, the L_2 distance between the EWM gradient of the remaining wild set $\mathcal{S}$ and the average InD gradient. For each candidate sample $i \in \mathcal{S}$, Medix evaluates the leave-one-out distance $\left\| \text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}} \right\|$ and assigns the reduction score $\delta_i = d_t - \left\| \text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}} \right\|$. The k samples with the largest reductions are removed from $\mathcal{S}$ and accumulated as candidate OOD points. The process continues until the maximum reduction is no longer significant or the iteration budget is exhausted. The threshold ϵ acts as an over-filtering safeguard: once additional removals yield only a small improvement in median-gradient distance, the algorithm stops to reduce the risk of discarding InD samples.

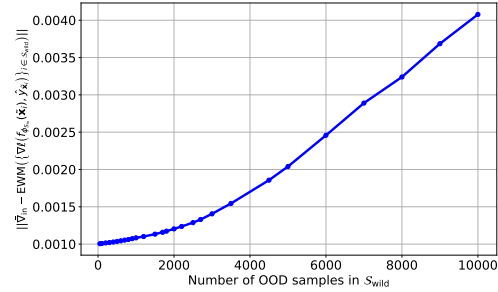


Figure 1: Gradient-median distance from the InD reference as OOD samples are added to $\mathcal{S}_{\text{wild}}$.

3.2 Training the OOD Detector with Candidate Outliers

After Medix identifies the candidate outlier set $\hat{\mathcal{S}}_{\text{out}}$, we train the detector g_{θ} to separate labeled InD samples from the extracted candidates, following Du et al. (2024a). The objective uses a zero-threshold decision rule: labeled InD samples $\mathbf{x} \in \mathcal{S}_{\text{in}}$ should receive positive detector outputs, whereas candidate outliers $\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}$ should receive negative outputs. The corresponding loss is

$$\begin{aligned} \mathcal{L}_{\mathcal{S}_{\text{in}}, \hat{\mathcal{S}}_{\text{out}}}(g_{\theta}) &= \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_{\theta}) + \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_{\theta}), \\ \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_{\theta}) &= \mathbb{E}_{\mathbf{x} \in \mathcal{S}_{\text{in}}} \mathbb{I}\{g_{\theta}(\mathbf{x}) \leq 0\}, \\ \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_{\theta}) &= \mathbb{E}_{\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}} \mathbb{I}\{g_{\theta}(\tilde{\mathbf{x}}) > 0\}. \end{aligned} \quad (5)$$

Algorithm 1 Iterative Outlier Detection via **Medix**

Require: $\bar{\nabla}_{\text{in}}, G_i = \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i})}$, maximum iterations T , hyperparameters ϵ, k

- 1: Initialize $\mathcal{S} \leftarrow \mathcal{S}_{\text{wild}}$ (wild set), $\mathcal{O} \leftarrow \emptyset$ (outliers), $t \leftarrow 0$ (iteration), $d_t \leftarrow 0$, $\delta_{\text{max}} \leftarrow \infty$ (deviations), $\mathcal{I}_k \leftarrow \emptyset$
- 2: **while** $t \leq T$ or $|\delta_{\text{max}}| > \epsilon$ **do**
- 3: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\tilde{\mathbf{x}}_i : i \in \mathcal{I}_k\}$ ▷ Add outliers to set $\mathcal{O}$
- 4: $d_t \leftarrow \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$ ▷ Compute L2 deviation
- 5: **for each** $i \in \mathcal{S}$ **do**
- 6: $\delta_i \leftarrow d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$
- 7: **end for**
- 8: $\mathcal{I}_k \leftarrow \text{indices}(\text{top-}k(\{\delta_i\}_{i \in \mathcal{S}}))$ ▷ Select top- k indices
- 9: $\mathcal{S} \leftarrow \mathcal{S} \setminus \mathcal{I}_k$ ▷ Remove outliers from $\mathcal{S}$
- 10: $\delta_{\text{max}} \leftarrow \max_{i \in \mathcal{S}} \{\delta_i\}$, $t \leftarrow t + 1$
- 11: **end while**
- 12: **return** $\mathcal{S}_{\text{out}} = \mathcal{O}$ ▷ Return detected outliers

171 Because the 0/1 loss is non-differentiable, optimization uses a differentiable sigmoid-based binary loss
 172 as a smooth surrogate for the same separation objective. We also retain the InD risk from Equation 2,
 173 so detector learning continues to preserve classification accuracy on InD samples. The resulting
 174 objective uses the extracted outlier candidates as explicit negative evidence while maintaining the
 175 supervised InD signal.

176 4 Theoretical Analysis

177 We now state the theoretical guarantees for the Medix filtering stage. The results analyze the
 178 median filtering objective by bounding the two errors induced by filtering an unlabeled mixture: InD
 179 samples incorrectly removed as outliers and OOD samples incorrectly retained as inliers. Together,
 180 the theorems characterize when the EWM criterion provides robust separation under the stated
 181 concentration, contamination, and separation assumptions. Full proofs appear in Appendix C.

Theorem 4.1 (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, and fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the EWM filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

182 Theorem 4.1 decomposes inlier misclassification into a *concentration term*, which decreases with
 183 sample size and captures finite-sample fluctuations of InD gradients, and a *contamination term*, which
 184 measures the influence of OOD gradients on the EWM. The bound remains controlled when $\pi < 0.5$,
 185 reflecting the median’s stability when fewer than half of the wild samples are OOD. We next bound
 186 the complementary error: true OOD samples that are not flagged and are therefore retained.

Theorem 4.2 (Outlier Misclassification Bound). *Assume that the gradients of OOD points in*

$\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ_{out}^2 . Suppose the mean OOD gradient μ_{out} satisfies a separation condition:

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d},$$

for some $\Delta > 0$. Then, for any tolerance $\epsilon \in (0, \Delta)$ and confidence level $\delta \in (0, 1)$, with probability at least $1 - \delta$, the outlier misclassification rate satisfies:

$$\text{ERR}_{\text{out}} \leq \underbrace{2d \exp\left(-\frac{(\Delta - \epsilon)^2}{2\sigma_{\text{out}}^2}\right)}_{\text{Separation term}} + \underbrace{\sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}}_{\text{Concentration term}} + \underbrace{\frac{1 - \pi}{2\pi}}_{\text{Contamination term}}.$$

188

189 Theorem 4.2 complements the inlier result by bounding the fraction of OOD points retained as InD.
 190 Together, Theorems 4.1 and 4.2 provide a two-sided robustness characterization of Medix filtering.
 191 The two misclassification rates are governed by three interpretable effects:

192 **Contamination Effect.** In Theorem 4.1, the term $\pi/[2(1 - \pi)]$ captures how OOD gradients can
 193 perturb the EWM; the median-based criterion remains stable as long as $\pi < 0.5$. Theorem 4.2
 194 contains the corresponding penalty $\frac{1-\pi}{2\pi}$, which reflects the difficulty of identifying OOD points when
 195 the OOD component is underrepresented in the mixture.

196 **Concentration Effect.** Both bounds use sub-Gaussian concentration of gradient coordinates. For
 197 InD data, this property keeps most gradients close to $\bar{\nabla}_{\text{in}}$ and stabilizes the median in finite samples.
 198 For OOD data, concentration around μ_{out} quantifies the probability that an OOD gradient falls close
 199 enough to the InD reference to evade filtering. These terms scale as $1/\sqrt{m}$ for the relevant sample
 200 count.

201 **Separation Effect.** Theorem 4.2 also requires gradient-level separation between OOD and InD
 202 behavior. The exponential term $\exp(-(\Delta - \epsilon)^2/2\sigma_{\text{out}}^2)$ formalizes this dependence: larger separation
 203 makes it less likely that OOD gradients are indistinguishable from the InD reference under the filter.

Theorems 4.1 and 4.2 jointly show that, with high probability, the median filtering objective can separate InD and OOD samples in contaminated unlabeled mixtures under the stated assumptions. The fraction of InD samples misclassified as outliers is bounded by contamination and concentration effects, while the fraction of OOD points incorrectly retained is governed by an exponential separation term, a concentration bound, and a reverse contamination effect. These results provide a formal explanation for the robustness observed in the implemented Medix filtering procedure.

204

205 **Remark 4.3. Sub-Gaussian Assumption for Gradient Coordinates.** The coordinate-wise sub-Gaussian
 206 assumption is used to obtain the concentration terms in the preceding bounds. Figure 4a provides
 207 qualitative empirical support for this assumption: the observed InD gradient histogram is bell-shaped
 208 and concentrated around the mean, suggesting relatively light tails in the examined setting. Figure
 209 4b further compares the empirical quantiles of the gradient distribution with Gaussian quantiles in a
 210 Q-Q plot (Wilk and Gnanadesikan, 1968); the approximate alignment with the 45-degree reference
 211 line is consistent with, though not a proof of, sub-Gaussian behavior. *We also provide a looser*
 212 *version of these bounds, which removes the sub-Gaussian assumption and holds under finite fourth*
 213 *moments. This version is presented in Theorem C.3 (Appendix C.3). The rates degrade, but the same*
 214 *median-contamination structure remains available under weaker tail assumptions.*

215 5 Experiments

216 This section evaluates Medix across the InD-OOD dataset pairs used in our benchmark and compares
 217 it with 20 widely used baselines. The experiments test whether median-based filtering improves OOD
 218 detection under the wild-data protocol adopted from prior work, while keeping explicit the data-access
 219 differences among baseline families. All experiments are performed on hardware equipped with
 220 NVIDIA A100-SXM4-80GB GPUs, and the code required to reproduce the results will be released
 221 upon acceptance.

5.1 Models, Datasets, and Baselines

Datasets. We follow the experimental protocol of Katz-Samuels et al. (2022a), which introduced learning OOD detectors from wild data, so that the main comparisons use an established evaluation setting. CIFAR-10 and CIFAR-100 serve as InD datasets ($\mathbb{P}_{\text{in}}$). OOD testing uses the natural-image datasets PLACES365 (Zhou et al., 2017), SVHN, TEXTURES (Cimpoi et al., 2014), and LSUN-RESIZE & LSUN-C (Yu et al., 2015) as OOD datasets ($\mathbb{P}_{\text{out}}$). To construct wild data ($\mathbb{P}_{\text{wild}}$), we mix a subset of InD samples ($\mathbb{P}_{\text{in}}$) with OOD samples ($\mathbb{P}_{\text{out}}$) using the default contamination parameter $\pi = 0.5$. For instance, when PLACES365 is the OOD test set, CIFAR samples are mixed with PLACES365 samples to form the wild set, and PLACES365 is also used as the OOD test distribution. This construction is repeated for each OOD dataset and each baseline, and the large-scale appendix experiment separately considers the case in which the wild OOD source and test OOD source differ. Each CIFAR InD dataset is split into two halves: the first 25,000 images train $\phi_{\mathcal{S}_{\text{in}}}$, and the remaining images contribute to $\mathcal{S}_{\text{wild}}$. Gradient features are computed from the penultimate-layer weights, following evidence that these gradients are informative for OOD detection (Huang et al., 2021a).

Evaluation metrics. We report three standard metrics: (1) False Positive Rate (FPR95 $\downarrow$), the rate at which OOD samples are classified as InD when the InD true positive rate is fixed at 95%; (2) Area Under the Receiver Operating Characteristic Curve (AUROC $\uparrow$); and (3) InD Classification Accuracy (InD Acc $\uparrow$).

Baselines. We organize baselines by data access. Among InD-only methods, we compare with Maximum Softmax Probability (MSP) (Hendrycks and Gimpel, 2016), ODIN (Liang et al., 2018), Mahalanobis Distance (Lee et al., 2018a), Energy Score (Liu et al., 2020b), ReAct (Sun et al., 2021), DICE (Sun and Li, 2022), KNN Distance (Sun et al., 2022), and ASH (Djurisic et al., 2022), all based on a model trained with softmax cross-entropy loss. We also include contrastive-loss methods CSI (Tack et al., 2020a) and KNN+ (Sun et al., 2022). Among methods that use InD and wild or auxiliary data, we compare with Outlier Exposure (OE) (Hendrycks et al., 2019), energy-regularization learning (Liu et al., 2020b), and WOODS (Katz-Samuels et al., 2022a), which uses wild unlabeled data through constrained optimization. Recent baselines CONJ (Peng et al., 2024) and DRL (Zhang et al., 2024) are included to broaden the comparison.

5.2 Experimental Setup

Following WOODS (Katz-Samuels et al., 2022a), the InD classifier $\phi_{\mathcal{S}_{\text{in}}}$ is a Wide ResNet (Zagoruyko, 2016) with 40 layers and width factor 2. It is trained by stochastic gradient descent with momentum 0.9, weight decay 0.0005, and initial learning rate 0.1 for 100 epochs, using cosine learning-rate decay, batch size 128, and dropout 0.3. Medix hyperparameters ϵ and k are selected from $\{5\text{e-}5, 5\text{e-}4, 5\text{e-}3, 5\text{e-}2\}$ and $\{4\text{k}, 7\text{k}, 10\text{k}, 20\text{k}\}$, respectively, accounting for dataset size and OOD performance as described in the sensitivity analysis. The OOD classifier g_θ is initialized from the pre-trained InD classifier $\phi_{\mathcal{S}_{\text{in}}}$ and augmented with a linear binary-classification layer on penultimate-layer features. We fine-tune this detector for 100 epochs with learning rate 0.001 according to Equation (5). The binary classification loss is combined with the InD classification loss using weight 10 on the binary component; all remaining training parameters match those used for $\phi_{\mathcal{S}_{\text{in}}}$.

5.3 Results

Table 2 reports the CIFAR-100 results and shows that Medix achieves the strongest OOD detection performance among the evaluated methods in this benchmark. Two patterns are especially salient. First, methods that use both InD and wild data generally outperform methods trained only on InD data, indicating the value of deployment-collected unlabeled samples under this protocol. Against KNN+, Medix reduces FPR95 by 52.31% on PLACES365 and 38.24% on TEXTURES. Second, Medix improves over methods that already use wild data: on CIFAR-100 it obtains an average FPR95 of 5.42%, a 1.32% improvement over WOODS, while maintaining an InD accuracy of 73.33%. The InD-accuracy comparison should be read in light of the training split, since Medix uses 25,000 labeled InD samples whereas baselines that do not leverage wild data use the full CIFAR-100 training set of 50,000 samples. Table 1 gives the corresponding CIFAR-10 results. Medix again improves over the baselines, including WOODS, with FPR95 gains of 7.21% on PLACES365 and 5.25% on TEXTURES.

Table 1: OOD detection performance with CIFAR-10 as InD data. The table reports Medix and baseline results averaged over five runs; best results are highlighted in **bold**.

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
Using $\mathbb{P}_m$ only													
MSP	48.49	91.89	59.48	88.20	30.80	95.65	52.15	91.37	59.28	88.50	50.04	91.12	94.84
ODIN	33.35	91.96	57.40	84.49	15.52	97.04	26.62	94.57	49.12	84.97	36.40	90.61	94.84
Mahalanobis	12.89	97.62	68.57	84.61	39.22	94.15	42.62	93.23	15.00	97.33	35.66	93.34	94.84
Energy	35.59	90.96	40.14	89.89	8.26	98.35	27.58	94.24	52.79	85.22	32.87	91.73	94.84
KNN	24.53	95.96	25.29	95.69	25.55	95.26	27.57	94.71	50.90	89.14	30.77	94.15	94.84
ReAct	89.57	81.44	90.44	90.44	14.38	97.21	33.63	93.58	53.63	86.59	36.77	91.48	94.84
DICE	35.44	89.65	46.83	86.69	6.32	98.68	28.93	93.56	53.62	82.20	34.23	90.16	94.84
ASH	6.51	98.65	48.45	88.34	0.90	99.73	4.96	98.92	24.34	95.09	17.03	96.15	94.84
CSI	17.30	97.40	34.95	93.64	1.95	99.55	12.15	98.01	20.45	95.93	17.36	96.91	94.17
KNN+	2.99	99.41	24.69	94.84	2.95	99.39	11.22	97.98	9.65	98.37	10.30	97.99	93.19
Using $\mathbb{P}_m$ and $\mathbb{P}_{wid}$													
OE	1.13	99.53	19.48	94.88	1.91	98.16	0.54	98.84	7.75	98.56	6.16	97.99	94.12
Energy (w/ OE)	5.24	98.72	14.66	96.18	2.35	99.30	4.85	98.62	10.51	97.10	7.52	97.98	94.24
WOODS	0.17	99.91	10.19	98.05	0.31	99.14	0.11	99.38	6.21	98.13	3.40	98.92	94.74
Medix	0.06	99.98	2.98	99.10	0.01	99.98	0.01	99.98	0.96	99.66	0.80	99.74	93.58
(Ours)	± 0.01	± 0.01	± 0.29	± 0.09	± 0.01	± 0.00	± 0.01	± 0.01	± 0.13	± 0.06	± 0.09	± 0.03	± 0.64

Table 2: OOD detection performance with CIFAR-100 as InD data. The table reports Medix and baseline results averaged over five runs; best results are highlighted in **bold**.

Methods	OOD Datasets												InD Acc $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
Using $\mathbb{P}_m$ only													
MSP	84.59	71.44	82.84	73.78	66.54	83.79	82.42	75.38	83.29	73.34	79.94	75.55	75.96
ODIN	84.66	67.26	87.88	71.63	55.55	87.73	71.96	81.82	79.27	73.45	75.86	76.38	75.96
Mahalanobis	57.52	86.01	88.83	67.87	91.18	69.69	21.23	96.00	39.39	90.57	59.63	82.03	75.96
Energy	85.82	73.99	80.56	75.44	35.32	93.53	79.47	79.23	79.41	76.28	72.12	79.69	75.96
KNN	66.38	83.76	79.17	71.91	70.96	83.71	77.83	78.85	88.00	67.19	76.47	77.08	75.96
ReAct	74.33	88.04	81.33	74.32	39.30	91.19	79.86	73.69	67.38	82.80	68.44	82.01	75.96
DICE	88.35	72.58	81.61	75.07	26.77	94.74	80.21	78.50	76.29	76.07	70.65	79.39	75.96
ASH	21.36	94.28	68.37	71.22	15.27	95.65	68.18	85.42	40.87	92.29	42.81	87.77	75.96
CSI	64.70	84.97	82.25	73.63	38.10	92.52	91.55	63.42	74.70	92.66	70.26	81.44	69.90
KNN+	32.21	93.74	68.30	75.31	40.37	86.13	44.86	88.88	46.26	87.40	46.40	86.29	73.78
Using $\mathbb{P}_m$ and $\mathbb{P}_{wild}$													
OE	2.86	99.05	40.21	88.75	4.13	99.05	1.25	99.38	22.86	94.63	14.26	96.17	73.38
Energy (w/ OE)	2.71	99.34	34.82	90.05	3.27	99.18	2.54	99.23	30.16	94.76	14.70	96.51	72.76
WOODS	0.17	99.80	21.87	93.73	0.48	99.61	1.24	99.54	9.95	95.97	6.74	97.73	73.91
Medix	0.16	99.96	15.99	95.23	0.13	99.98	0.83	99.83	8.02	97.79	5.42	98.96	73.33
(Ours)	± 0.02	± 0.00	± 0.66	± 0.14	± 0.06	± 0.02	± 0.36	± 0.06	± 0.75	± 0.30	± 0.37	± 0.10	± 0.83

A representative visual example of Medix. We further examine the outlier-extraction behavior of Algorithm 1 using a two-dimensional synthetic example. This simplified setting makes the filtering decision visually interpretable. InD samples are drawn from three multivariate Gaussian distributions corresponding to three classes, with means $[-2, 0]$, $[2, 0]$, and $[0, 2\sqrt{3}]$, and shared covariance $0.25 \cdot I$. OOD samples are drawn from $\mathcal{N}([20, 2\sqrt{3}], 0.25 \cdot I)$. Figure 2 shows that the outliers identified by Medix largely coincide with the ground-truth OOD region; quantitatively, Medix flags 87.5% of actual OOD samples as outliers.

Additional studies and insights. Appendix A reports additional analyses that delineate the empirical scope of the method: hyperparameter selection and sensitivity (Appendix A.2); a comparison between EWM and geometric median showing that EWM is more responsive to distributional shifts in this filtering procedure (Appendix A.1); a comprehensive comparison with competitive contrastive-learning baselines (Appendix A.3); evaluation under a large-scale unseen OOD setting with $\mathbb{P}_{out}^{test} \neq \mathbb{P}_{out}$ (Appendix A.4); computation and memory profiling (Appendix A.6); the effect of pseudo-label confidence (Appendix A.5); and a comparison with semi-supervised open-set recognition methods (Appendix A.7).

6 Related Work

OOD detection has become a substantial research area because reliable prediction under distribution shift requires mechanisms for recognizing inputs outside the training distribution (Fort et al., 2021; Yang et al., 2024; Fang et al., 2022; Zhu et al., 2022; Yang et al., 2022; Wang et al., 2022c; Galil et al., 2023; Djuricic et al., 2023; Tao et al., 2023; Zheng et al., 2023; Wang et al., 2022b, 2023b; Uppaal et al., 2023; Zhu et al., 2023; Bai et al., 2023; Ming and Li, 2024; Zhang et al., 2023; Ghosal et al., 2024; Abbas et al., 2025). One major line of work assigns OOD scores to inputs, including distance-based methods (Lee et al., 2018b; Tack et al., 2020b; Ren et al., 2021; Schwag et al., 2021; Sun et al., 2022; Du et al., 2022a; Ming et al., 2023; Ren et al., 2022), gradient-based scores (Huang et al., 2021b), energy-based scores (Liu et al., 2020c; Wang et al., 2021; Wu et al., 2023), confidence-

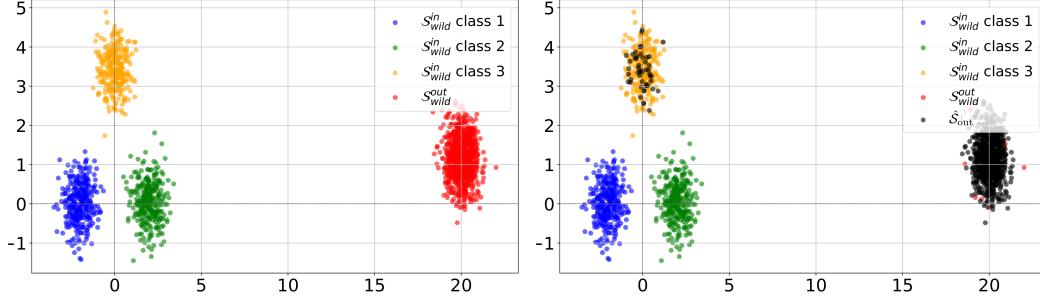


Figure 2: Medix on unlabeled synthetic wild data. (a) InD data S_{wild}^{in} and OOD data S_{wild}^{out} in the wild mixture, with inliers sampled from three multivariate Gaussian distributions. (b) Outliers $\hat{S}_{out}$ filtered by Medix (black); 12.5% of $\hat{S}_{out}$ consists of InD data S_{wild}^{in} .

based approaches (Bendale and Boulton, 2016; Hendrycks and Gimpel, 2017; Liang et al., 2018), and Bayesian methods (Gal and Ghahramani, 2016; Lakshminarayanan et al., 2017; Maddox et al., 2019; Malinin and Gales, 2019; Wen et al., 2020; Kristiadi et al., 2020).

A second line modifies training through regularization (Malinin and Gales, 2018; Geifman and El-Yaniv, 2019; Hein et al., 2019; Meinke and Hein, 2020; Jeong and Kim, 2020; Liu et al., 2020a; Van Amersfoort et al., 2020; Yang et al., 2021; Wei et al., 2022; Du et al., 2022b, 2023; Wang et al., 2023a). These methods often reduce confidence (Lee et al., 2017; Hendrycks et al., 2019) or increase energy (Liu et al., 2020c; Du et al., 2022c; Ming et al., 2022) on OOD data, but many require access to an auxiliary OOD dataset.

Several studies (Zhou et al., 2021; Katz-Samuels et al., 2022b; He et al., 2023) relax the closed labeled-data assumption by incorporating unlabeled wild data (Katz-Samuels et al., 2022a; Geng et al., 2025), although they do not provide the same explicit median-based mechanism for outlier extraction. Du et al. (2024a,b) introduce explicit filtering, but their thresholding technique differs from the median-centric criterion developed here. In addition, Katz-Samuels et al. (2022a); Du et al. (2024a) assume batch-level mixing, where each batch has a specified InD/OOD ratio. In large outsourced datasets, such batch structure is often unavailable because samples are mixed randomly across the dataset; Medix instead operates at the dataset level without requiring structured batches. Positive-unlabeled learning methods also train classifiers from positive and/or unlabeled data (Letouzey et al., 2000; Hsieh et al., 2015; Plessis et al., 2015; Niu et al., 2016; Gong et al., 2018; Chapel et al., 2020; Garg et al., 2021; Xu and Denil, 2021; Garg et al., 2022; Zhao et al., 2022; Acharya et al., 2022); however, these approaches primarily distinguish $\mathbb{P}_{out}$ from $\mathbb{P}_{in}$ rather than combining median-based outlier identification with OOD detector training and provable filtering error bounds.

7 Conclusions

We introduced Medix, a median-centric framework for OOD detection from unlabeled in-the-wild data. By using the robustness of the median in gradient space, Medix filters candidate outliers from mixed unlabeled data and uses them to train an OOD detector while preserving the InD classification objective. The theoretical analysis provides high-probability bounds on inlier misclassification under contamination up to 50%, with errors controlled by concentration and contamination terms, and gives a complementary bound for OOD retention under gradient separation. Across the evaluated benchmarks, Medix improves over 20 baselines, reducing average FPR95 by 40.98% relative to KNN+ on CIFAR-100, outperforming methods such as WOODS and DRL in the reported comparisons, and achieving an outlier extraction error rate as low as 12.5% in the synthetic visualization. These results support the central claim that median-based filtering is an effective and theoretically motivated mechanism for using unlabeled wild data in OOD detection under the studied protocols.

References

Abbas, M., Azmat, M., Horesh, R., and Yurochkin, M. (2025). Out-of-distribution detection using synthetic data generation. In *Second Conference on Language Modeling*.

336 Acharya, A., Dhillon, I. S., and Sanghavi, S. (2024). Geometric median (gm) matching for robust data pruning.
337 *arXiv preprint arXiv:2406.17188*.

338 Acharya, A., Sanghavi, S., Jing, L., Bhushanam, B., Choudhary, D., Rabbat, M., and Dhillon, I. (2022). Positive
339 unlabeled contrastive learning. *arXiv preprint arXiv:2206.01206*.

340 Bai, H., Canal, G., Du, X., Kwon, J., Nowak, R. D., and Li, Y. (2023). Feed two birds with one scone: Exploiting
341 wild data for both out-of-distribution generalization and detection. In *International Conference on Machine*
342 *Learning*, pages 1454–1471. PMLR.

343 Bendale, A. and Boulton, T. E. (2016). Towards open set deep networks. In *Proceedings of the IEEE conference*
344 *on computer vision and pattern recognition*, pages 1563–1572.

345 Chapel, L., Alaya, M. Z., and Gasso, G. (2020). Partial optimal transport with applications on positive-unlabeled
346 learning. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural*
347 *Information Processing Systems*, volume 33, pages 2903–2913. Curran Associates, Inc.

348 Cimpoi, M., Maji, S., Kokkinos, I., Mohamed, S., and Vedaldi, A. (2014). Describing textures in the wild. In
349 *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3606–3613.

350 Djurisic, A., Bozanic, N., Ashok, A., and Liu, R. (2022). Extremely simple activation shaping for out-of-
351 distribution detection. *arXiv preprint arXiv:2209.09858*.

352 Djurisic, A., Bozanic, N., Ashok, A., and Liu, R. (2023). Extremely simple activation shaping for out-of-
353 distribution detection. In *The Eleventh International Conference on Learning Representations*.

354 Du, X., Fang, Z., Diakonikolas, I., and Li, Y. (2024a). How does unlabeled data provably help out-of-distribution
355 detection? In *The Twelfth International Conference on Learning Representations*.

356 Du, X., Ghosh, R., Sim, R., Salem, A., Carvalho, V., Lawton, E., Li, Y., and Stokes, J. W. (2024b). Vlmguard:
357 Defending vlms against malicious prompts via unlabeled data. *arXiv preprint arXiv:2410.00296*.

358 Du, X., Gozum, G., Ming, Y., and Li, Y. (2022a). Siren: Shaping representations for detecting out-of-distribution
359 objects. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A., editors, *Advances in*
360 *Neural Information Processing Systems*, volume 35, pages 20434–20449. Curran Associates, Inc.

361 Du, X., Sun, Y., Zhu, J., and Li, Y. (2023). Dream the impossible: Outlier imagination with diffusion models.
362 In Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., and Levine, S., editors, *Advances in Neural*
363 *Information Processing Systems*, volume 36, pages 60878–60901. Curran Associates, Inc.

364 Du, X., Wang, X., Gozum, G., and Li, Y. (2022b). Unknown-aware object detection: Learning what you don’t
365 know from videos in the wild. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern*
366 *Recognition*, pages 13678–13688.

367 Du, X., Wang, Z., Cai, M., and Li, S. (2022c). Towards unknown-aware learning with virtual outlier synthesis.
368 In *International Conference on Learning Representations*.

369 Fan, Y., Kukleva, A., Dai, D., and Schiele, B. (2023). Ssb: Simple but strong baseline for boosting performance
370 of open-set semi-supervised learning. In *Proceedings of the IEEE/CVF International Conference on Computer*
371 *Vision (ICCV)*, pages 16068–16078.

372 Fang, Z., Li, Y., Lu, J., Dong, J., Han, B., and Liu, F. (2022). Is out-of-distribution detection learnable? In
373 Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A., editors, *Advances in Neural*
374 *Information Processing Systems*, volume 35, pages 37199–37213. Curran Associates, Inc.

375 Fort, S., Ren, J., and Lakshminarayanan, B. (2021). Exploring the limits of out-of-distribution detection.
376 *Advances in Neural Information Processing Systems*, 34:7068–7081.

377 Gal, Y. and Ghahramani, Z. (2016). Dropout as a bayesian approximation: Representing model uncertainty
378 in deep learning. In Balcan, M. F. and Weinberger, K. Q., editors, *Proceedings of The 33rd International*
379 *Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 1050–
380 1059, New York, New York, USA. PMLR.

381 Galil, I., Dabbah, M., and El-Yaniv, R. (2023). A framework for benchmarking class-out-of-distribution detection
382 and its application to imagenet. In *The Eleventh International Conference on Learning Representations*.

383 Garg, S., Balakrishnan, S., and Lipton, Z. (2022). Domain adaptation under open set label shift. In Koyejo,
384 S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A., editors, *Advances in Neural Information*
385 *Processing Systems*, volume 35, pages 22531–22546. Curran Associates, Inc.

386 Garg, S., Wu, Y., Smola, A. J., Balakrishnan, S., and Lipton, Z. (2021). Mixture proportion estimation and
387 pu learning: a modern approach. In Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan,
388 J. W., editors, *Advances in Neural Information Processing Systems*, volume 34, pages 8532–8544. Curran
389 Associates, Inc.

390 Geifman, Y. and El-Yaniv, R. (2019). SelectiveNet: A deep neural network with an integrated reject option. In
391 Chaudhuri, K. and Salakhutdinov, R., editors, *Proceedings of the 36th International Conference on Machine
392 Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2151–2159. PMLR.

393 Geng, C., Li, Q., Wang, X., Liang, D., Chen, S., and Yuen, P. C. (2025). Lod: Loss-difference ood detection by
394 intentionally label-noisifying unlabeled wild data. *arXiv preprint arXiv:2505.12952*.

395 Ghosal, S. S., Sun, Y., and Li, Y. (2024). How to overcome curse-of-dimensionality for out-of-distribution
396 detection? In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 19849–19857.

397 Gong, T., Wang, G., Ye, J., Xu, Z., and Lin, M. (2018). Margin based pu learning. In *Proceedings of the AAAI
398 Conference on Artificial Intelligence*, volume 32.

399 Hang, J.-Y. and Zhang, M.-L. (2024). Binary decomposition: A problem transformation perspective for open-set
400 semi-supervised learning. In Salakhutdinov, R., Koltner, Z., Heller, K., Weller, A., Oliver, N., Scarlett, J., and
401 Berkenkamp, F., editors, *Proceedings of the 41st International Conference on Machine Learning*, volume 235
402 of *Proceedings of Machine Learning Research*, pages 17505–17518. PMLR.

403 He, R., Li, R., Han, Z., Yang, X., and Yin, Y. (2023). Topological structure learning for weakly-supervised
404 out-of-distribution detection. In *Proceedings of the 31st ACM International Conference on Multimedia*, pages
405 4858–4866.

406 Hein, M., Andriushchenko, M., and Bitterwolf, J. (2019). Why relu networks yield high-confidence predictions
407 far away from the training data and how to mitigate the problem. In *Proceedings of the IEEE/CVF conference
408 on computer vision and pattern recognition*, pages 41–50.

409 Hendrycks, D. and Gimpel, K. (2016). A baseline for detecting misclassified and out-of-distribution examples in
410 neural networks. *arXiv preprint arXiv:1610.02136*.

411 Hendrycks, D. and Gimpel, K. (2017). A baseline for detecting misclassified and out-of-distribution examples in
412 neural networks. In *International Conference on Learning Representations*.

413 Hendrycks, D., Mazeika, M., and Dietterich, T. (2019). Deep anomaly detection with outlier exposure. In
414 *International Conference on Learning Representations*.

415 Hsieh, C.-J., Natarajan, N., and Dhillon, I. (2015). Pu learning for matrix completion. In Bach, F. and Blei, D.,
416 editors, *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings
417 of Machine Learning Research*, pages 2445–2453, Lille, France. PMLR.

418 Huang, R., Geng, A., and Li, Y. (2021a). On the importance of gradients for detecting distributional shifts in the
419 wild. *Advances in Neural Information Processing Systems*, 34:677–689.

420 Huang, R., Geng, A., and Li, Y. (2021b). On the importance of gradients for detecting distributional shifts in the
421 wild. In Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan, J. W., editors, *Advances in Neural Information
422 Processing Systems*.

423 Huber, P. J. (1964). Robust estimation of a location parameter. *The Annals of Mathematical Statistics*, 35(1):73–
424 101.

425 Jeong, T. and Kim, H. (2020). Ood-maml: Meta-learning for few-shot out-of-distribution detection and
426 classification. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in
427 Neural Information Processing Systems*, volume 33, pages 3907–3916. Curran Associates, Inc.

428 Katz-Samuels, J., Nakhleh, J. B., Nowak, R., and Li, Y. (2022a). Training ood detectors in their natural habitats.
429 In *International Conference on Machine Learning*, pages 10848–10865. PMLR.

430 Katz-Samuels, J., Nakhleh, J. B., Nowak, R., and Li, Y. (2022b). Training OOD detectors in their natural
431 habitats. In Chaudhuri, K., Jegelka, S., Song, L., Szepesvari, C., Niu, G., and Sabato, S., editors, *Proceedings
432 of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning
433 Research*, pages 10848–10865. PMLR.

434 Kim, S., Kim, D., Cho, M., and Kwak, S. (2020). Proxy anchor loss for deep metric learning. In *2020 IEEE/CVF
435 Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3235–3244.

436 Kristiadi, A., Hein, M., and Hennig, P. (2020). Being bayesian, even just a bit, fixes overconfidence in relu
437 networks. In *International conference on machine learning*, pages 5436–5446. PMLR.

438 Krizhevsky, A., Hinton, G., et al. (2009). Learning multiple layers of features from tiny images.

439 Lakshminarayanan, B., Pritzel, A., and Blundell, C. (2017). Simple and scalable predictive uncertainty estimation
440 using deep ensembles. In Guyon, I., Luxburg, U. V., Bengio, S., Wallach, H., Fergus, R., Vishwanathan, S.,
441 and Garnett, R., editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates,
442 Inc.

443 Lee, K., Lee, H., Lee, K., and Shin, J. (2017). Training confidence-calibrated classifiers for detecting out-of-
444 distribution samples. *arXiv preprint arXiv:1711.09325*.

445 Lee, K., Lee, K., Lee, H., and Shin, J. (2018a). A simple unified framework for detecting out-of-distribution
446 samples and adversarial attacks. *Advances in neural information processing systems*, 31.

447 Lee, K., Lee, K., Lee, H., and Shin, J. (2018b). A simple unified framework for detecting out-of-distribution
448 samples and adversarial attacks. In Bengio, S., Wallach, H., Larochelle, H., Grauman, K., Cesa-Bianchi, N.,
449 and Garnett, R., editors, *Advances in Neural Information Processing Systems*, volume 31. Curran Associates,
450 Inc.

451 Letouzey, F., Denis, F., and Gilleron, R. (2000). Learning from positive and unlabeled examples. In *International
452 Conference on Algorithmic Learning Theory*, pages 71–85. Springer.

453 Liang, S., Li, Y., and Srikant, R. (2018). Enhancing the reliability of out-of-distribution image detection in
454 neural networks. In *International Conference on Learning Representations*.

455 Liu, J., Lin, Z., Padhy, S., Tran, D., Bedrax Weiss, T., and Lakshminarayanan, B. (2020a). Simple and principled
456 uncertainty estimation with deterministic deep learning via distance awareness. In Larochelle, H., Ranzato,
457 M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural Information Processing Systems*,
458 volume 33, pages 7498–7512. Curran Associates, Inc.

459 Liu, W., Wang, X., Owens, J., and Li, Y. (2020b). Energy-based out-of-distribution detection. *Advances in
460 neural information processing systems*, 33:21464–21475.

461 Liu, W., Wang, X., Owens, J., and Li, Y. (2020c). Energy-based out-of-distribution detection. In Larochelle,
462 H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in Neural Information Processing
463 Systems*, volume 33, pages 21464–21475. Curran Associates, Inc.

464 Maddox, W. J., Izmailov, P., Garipov, T., Vetrov, D. P., and Wilson, A. G. (2019). A simple baseline for bayesian
465 uncertainty in deep learning. In Wallach, H., Larochelle, H., Beygelzimer, A., d’Alché-Buc, F., Fox, E., and
466 Garnett, R., editors, *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc.

467 Malinin, A. and Gales, M. (2018). Predictive uncertainty estimation via prior networks. In Bengio, S., Wallach,
468 H., Larochelle, H., Grauman, K., Cesa-Bianchi, N., and Garnett, R., editors, *Advances in Neural Information
469 Processing Systems*, volume 31. Curran Associates, Inc.

470 Malinin, A. and Gales, M. (2019). Reverse kl-divergence training of prior networks: Improved uncertainty and
471 adversarial robustness. In Wallach, H., Larochelle, H., Beygelzimer, A., d’Alché-Buc, F., Fox, E., and Garnett,
472 R., editors, *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc.

473 Meinke, A. and Hein, M. (2020). Towards neural networks that provably know when they don’t know. In
474 *International Conference on Learning Representations*.

475 Ming, Y., Fan, Y., and Li, Y. (2022). POEM: Out-of-distribution detection with posterior sampling. In
476 Chaudhuri, K., Jegelka, S., Song, L., Szepesvari, C., Niu, G., and Sabato, S., editors, *Proceedings of the 39th
477 International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*,
478 pages 15650–15665. PMLR.

479 Ming, Y. and Li, Y. (2024). How does fine-tuning impact out-of-distribution detection for vision-language
480 models? *International Journal of Computer Vision*, 132(2):596–609.

481 Ming, Y., Sun, Y., Dia, O., and Li, Y. (2023). How to exploit hyperspherical embeddings for out-of-distribution
482 detection? In *The Eleventh International Conference on Learning Representations*.

483 Netzer, Y., Wang, T., Coates, A., Bissacco, A., Wu, B., Ng, A. Y., et al. (2011). Reading digits in natural images
484 with unsupervised feature learning. In *NIPS workshop on deep learning and unsupervised feature learning*,
485 volume 2011, page 4. Granada.

486 Nguyen, A., Yosinski, J., and Clune, J. (2015). Deep neural networks are easily fooled: High confidence
487 predictions for unrecognizable images. In *Proceedings of the IEEE conference on computer vision and pattern*
488 *recognition*, pages 427–436.

489 Niu, G., Du Plessis, M. C., Sakai, T., Ma, Y., and Sugiyama, M. (2016). Theoretical comparisons of positive-
490 unlabeled learning against positive-negative learning. *Advances in neural information processing systems*,
491 29.

492 Peng, B., Luo, Y., Zhang, Y., Li, Y., and Fang, Z. (2024). Conjnorm: Tractable density estimation for out-of-
493 distribution detection. In *The Twelfth International Conference on Learning Representations*.

494 Plessis, M. D., Niu, G., and Sugiyama, M. (2015). Convex formulation for learning from positive and unlabeled
495 data. In Bach, F. and Blei, D., editors, *Proceedings of the 32nd International Conference on Machine Learning*,
496 volume 37 of *Proceedings of Machine Learning Research*, pages 1386–1394, Lille, France. PMLR.

497 Ren, J., Fort, S., Liu, J., Roy, A. G., Padhy, S., and Lakshminarayanan, B. (2021). A simple fix to mahalanobis
498 distance for improving near-ood detection. *arXiv preprint arXiv:2106.09022*.

499 Ren, J., Luo, J., Zhao, Y., Krishna, K., Saleh, M., Lakshminarayanan, B., and Liu, P. J. (2022). Out-of-distribution
500 detection and selective generation for conditional language models. In *The Eleventh International Conference*
501 *on Learning Representations*.

502 Saito, K., Kim, D., and Saenko, K. (2021). Openmatch: Open-set semi-supervised learning with open-set
503 consistency regularization. In Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan, J. W.,
504 editors, *Advances in Neural Information Processing Systems*, volume 34, pages 25956–25967. Curran
505 Associates, Inc.

506 Sehwag, V., Chiang, M., and Mittal, P. (2021). {SSD}: A unified framework for self-supervised outlier detection.
507 In *International Conference on Learning Representations*.

508 Sun, Y., Guo, C., and Li, Y. (2021). React: Out-of-distribution detection with rectified activations. *Advances in*
509 *Neural Information Processing Systems*, 34:144–157.

510 Sun, Y. and Li, Y. (2022). Dice: Leveraging sparsification for out-of-distribution detection. In *European*
511 *Conference on Computer Vision*, pages 691–708. Springer.

512 Sun, Y., Ming, Y., Zhu, X., and Li, Y. (2022). Out-of-distribution detection with deep nearest neighbors. In
513 *International Conference on Machine Learning*, pages 20827–20840. PMLR.

514 Tack, J., Mo, S., Jeong, J., and Shin, J. (2020a). Csi: Novelty detection via contrastive learning on distributionally
515 shifted instances. *Advances in neural information processing systems*, 33:11839–11852.

516 Tack, J., Mo, S., Jeong, J., and Shin, J. (2020b). Csi: Novelty detection via contrastive learning on distributionally
517 shifted instances. In Larochelle, H., Ranzato, M., Hadsell, R., Balcan, M., and Lin, H., editors, *Advances in*
518 *Neural Information Processing Systems*, volume 33, pages 11839–11852. Curran Associates, Inc.

519 Tao, L., Du, X., Zhu, J., and Li, Y. (2023). Non-parametric outlier synthesis. In *The Eleventh International*
520 *Conference on Learning Representations*.

521 Uppaal, R., Hu, J., and Li, Y. (2023). Is fine-tuning needed? pre-trained language models are near perfect
522 for out-of-domain detection. In Rogers, A., Boyd-Graber, J., and Okazaki, N., editors, *Proceedings of the*
523 *61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages
524 12813–12832, Toronto, Canada. Association for Computational Linguistics.

525 Van Amersfoort, J., Smith, L., Teh, Y. W., and Gal, Y. (2020). Uncertainty estimation using a single deep
526 deterministic neural network. In *International conference on machine learning*, pages 9690–9700. PMLR.

527 Wang, H., Li, Z., Feng, L., and Zhang, W. (2022a). Vim: Out-of-distribution with virtual-logit matching. In
528 *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 4921–4930.

529 Wang, H., Liu, W., Bocchieri, A., and Li, Y. (2021). Can multi-label classification networks know what they
530 don’t know? *Advances in Neural Information Processing Systems*, 34:29074–29087.

531 Wang, Q., Fang, Z., Zhang, Y., Liu, F., Li, Y., and Han, B. (2023a). Learning to augment distributions for
532 out-of-distribution detection. In Oh, A., Naumann, T., Globerson, A., Saenko, K., Hardt, M., and Levine,
533 S., editors, *Advances in Neural Information Processing Systems*, volume 36, pages 73274–73286. Curran
534 Associates, Inc.

Wang, Q., Liu, F., Zhang, Y., Zhang, J., Gong, C., Liu, T., and Han, B. (2022b). Watermarking for out-of-distribution detection. *Advances in Neural Information Processing Systems*, 35:15545–15557.

Wang, Q., Ye, J., Liu, F., Dai, Q., Kalander, M., Liu, T., HAO, J., and Han, B. (2023b). Out-of-distribution detection with implicit outlier transformation. In *The Eleventh International Conference on Learning Representations*.

Wang, Y., Zou, J., Lin, J., Ling, Q., Pan, Y., Yao, T., and Mei, T. (2022c). Out-of-distribution detection via conditional kernel independence model. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., and Oh, A., editors, *Advances in Neural Information Processing Systems*, volume 35, pages 36411–36425. Curran Associates, Inc.

Wang, Z., Xiang, L., Huang, L., Mao, J., Xiao, L., and Yamasaki, T. (2024). Scomatch: Alleviating overtrusting in open-set semi-supervised learning. In *European Conference on Computer Vision*, pages 217–233. Springer.

Wei, H., Xie, R., Cheng, H., Feng, L., An, B., and Li, Y. (2022). Mitigating neural network overconfidence with logit normalization. In Chaudhuri, K., Jegelka, S., Song, L., Szepesvari, C., Niu, G., and Sabato, S., editors, *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 23631–23644. PMLR.

Wen, Y., Tran, D., and Ba, J. (2020). Batchensemble: an alternative approach to efficient ensemble and lifelong learning. In *International Conference on Learning Representations*.

Wilk, M. B. and Gnanadesikan, R. (1968). Probability plotting methods for the analysis of data. *Biometrika*, 55(1):1–17.

Wu, Q., Chen, Y., Yang, C., and Yan, J. (2023). Energy-based out-of-distribution detection for graph neural networks. In *The Eleventh International Conference on Learning Representations*.

Xu, D. and Denil, M. (2021). Positive-unlabeled reward learning. In *Conference on Robot Learning*, pages 205–219. PMLR.

Yang, J., Wang, H., Feng, L., Yan, X., Zheng, H., Zhang, W., and Liu, Z. (2021). Semantically coherent out-of-distribution detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 8301–8309.

Yang, J., Wang, P., Zou, D., Zhou, Z., Ding, K., Peng, W., Wang, H., Chen, G., Li, B., Sun, Y., et al. (2022). Openood: Benchmarking generalized out-of-distribution detection. *Advances in Neural Information Processing Systems*, 35:32598–32611.

Yang, J., Zhou, K., Li, Y., and Liu, Z. (2024). Generalized out-of-distribution detection: A survey. *International Journal of Computer Vision*, 132(12):5635–5662.

Yu, F., Seff, A., Zhang, Y., Song, S., Funkhouser, T., and Xiao, J. (2015). Lsun: Construction of a large-scale image dataset using deep learning with humans in the loop. *arXiv preprint arXiv:1506.03365*.

Zagoruyko, S. (2016). Wide residual networks. *arXiv preprint arXiv:1605.07146*.

Zhang, J., Yang, J., Wang, P., Wang, H., Lin, Y., Zhang, H., Sun, Y., Du, X., Zhou, K., Zhang, W., et al. (2023). Openood v1. 5: Enhanced benchmark for out-of-distribution detection. *arXiv preprint arXiv:2306.09301*.

Zhang, Y., Lu, J., Peng, B., Fang, Z., and ming Cheung, Y. (2024). Learning to shape in-distribution feature space for out-of-distribution detection. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*.

Zhao, Y., Xu, Q., Jiang, Y., Wen, P., and Huang, Q. (2022). Dist-pu: Positive-unlabeled learning from a label distribution perspective. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14461–14470.

Zheng, H., Wang, Q., Fang, Z., Xia, X., Liu, F., Liu, T., and Han, B. (2023). Out-of-distribution detection learning with unreliable out-of-distribution sources. *Advances in Neural Information Processing Systems*, 36:72110–72123.

Zhou, B., Lapedriza, A., Khosla, A., Oliva, A., and Torralba, A. (2017). Places: A 10 million image database for scene recognition. *IEEE transactions on pattern analysis and machine intelligence*, 40(6):1452–1464.

Zhou, Z., Guo, L.-Z., Cheng, Z., Li, Y.-F., and Pu, S. (2021). Step: Out-of-distribution detection in the presence of limited in-distribution labeled data. In Ranzato, M., Beygelzimer, A., Dauphin, Y., Liang, P., and Vaughan, J. W., editors, *Advances in Neural Information Processing Systems*, volume 34, pages 29168–29180. Curran Associates, Inc.

- 586 Zhu, J., Geng, Y., Yao, J., Liu, T., Niu, G., Sugiyama, M., and Han, B. (2023). Diversified outlier exposure
587 for out-of-distribution detection via informative extrapolation. *Advances in Neural Information Processing*
588 *Systems*, 36:22702–22734.
- 589 Zhu, Y., Chen, Y., Xie, C., Li, X., Zhang, R., Xue', H., Tian, X., zheng, b., and Chen, Y. (2022). Boosting
590 out-of-distribution detection with typical features. In Koyejo, S., Mohamed, S., Agarwal, A., Belgrave,
591 D., Cho, K., and Oh, A., editors, *Advances in Neural Information Processing Systems*, volume 35, pages
592 20758–20769. Curran Associates, Inc.

593 A Additional Studies

594 A.1 Element-wise Median vs Geometric Median

595 A natural design question is why Medix uses the element-wise median rather
 596 than the geometric median (Acharya et al., 2024) for outlier extraction.
 597 We study this design choice with CIFAR-10 as the InD
 598 dataset and SVHN as the OOD dataset. The wild set $\mathcal{S}_{\text{wild}}$
 599 initially contains 5k CIFAR-10 samples disjoint from the
 600 training set used to learn $\phi_{\mathcal{S}_{\text{in}}}$, which is then used to com-
 601 pute $\bar{\nabla}_{\text{in}}$. We incrementally add SVHN OOD samples and
 602 track both the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the
 603 median gradient statistic of the wild set and the proportion
 604 of OOD samples removed. Figure 3 shows that, for the
 605 same number of OOD samples in the wild set, the element-
 606 wise median identifies a substantially larger proportion
 607 of OOD samples as outliers than the geometric median.
 608 In this filtering procedure, EWM is more responsive to
 609 distributional shifts and is therefore the median statistic
 610 used in Medix.

Table 3: Sensitivity of Medix to filtering hyperparameters ϵ and k .

Method	FPR95↓	ϵ	k
DICE	88.35	—	—
ASH	21.36	—	—
CSI	64.70	—	—
KNN+	32.21	—	—
OE	2.86	—	—
Energy	2.71	—	—
Medix	0.16	0.005	20000
Medix	0.20	0.0005	20000
Medix	0.68	0.005	10000

611 A.2 Hyperparameter Selection and Sensitivity Analysis

612 We assess sensitivity to the filtering threshold ϵ and removal parameter k using CIFAR-100 as the InD
 613 dataset and SVHN as the OOD dataset. Table 3 shows that Medix remains effective across the tested
 614 settings. With $\epsilon = 0.005$ and $k = 20000$, Medix achieves FPR95 of 0.16. Reducing ϵ to 0.0005 or
 615 halving k to 10000 yields FPR95 values of 0.20 and 0.68, respectively, indicating gradual rather than
 616 abrupt degradation in this ablation. Thus, hyperparameter selection affects the operating point, but
 617 the tested choices preserve strong detection performance relative to the listed baselines.

618 A.3 Comprehensive comparison with recent competitive methods

619 We extend the comparison to competitive baselines with diverse contrastive-learning objectives,
 620 including SSD+ (Sehwag et al., 2021), ProxyAnchor (Kim et al., 2020), and CIDER Ming et al. (2023).
 621 Although Medix does not use a contrastive objective, these methods provide a strong benchmark
 622 for assessing whether the median-filtering mechanism remains competitive against representation-
 623 learning approaches. We also include recent methods CONJ (Peng et al., 2024), DRL (Zhang et al.,
 624 2024), Vim (Wang et al., 2022a), and VOS (Du et al., 2022c). For this experiment, CIFAR-100
 625 is the InD dataset and the ResNet-34 model¹ follows the setup in Ming et al. (2023); Zhang et al.
 626 (2024). The model is trained with stochastic gradient descent, momentum 0.9, and weight decay
 627 10^{-4} for 500 epochs, using initial learning rate 0.5 with cosine scheduling and batch size 512. We
 628 use the checkpoints provided by Ming et al. (2023)². Table 4 shows that Medix remains strongest in

¹Since ResNet-34 is the standard model across these works, we adopt the same model to ensure consistency and facilitate a fair comparison.

²<https://github.com>

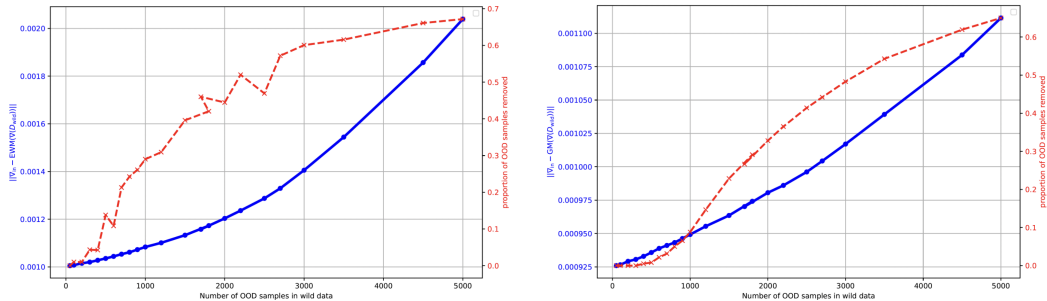


Figure 3: Comparison of element-wise median (EWM) and geometric median (GM).

Table 4: OOD detection performance with CIFAR-100 as InD data against recent competitive baselines. Values are averaged over five runs; best results are highlighted in **bold**.

Methods	OOD Datasets									
	SVHN		PLACES365		LSUN-C		TEXTURES		AVERAGE	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MSP	78.89	79.80	84.38	74.21	83.47	75.28	86.51	72.53	83.31	75.46
Mahalanobis	87.09	80.62	84.63	73.89	84.15	79.43	61.72	84.87	79.40	79.70
ODIN	70.16	84.88	82.16	75.19	76.36	80.10	85.28	75.23	78.49	78.85
Energy	66.91	85.25	81.41	76.37	59.77	86.69	79.01	79.96	71.78	82.07
ReAct	50.93	88.75	83.55	73.10	64.02	80.31	64.40	81.95	65.72	81.03
KNN	46.25	90.39	82.08	75.44	60.85	85.61	62.39	83.95	62.89	83.85
Vim	73.42	84.62	85.34	69.34	86.96	69.74	74.56	76.23	80.07	74.98
VOS	43.24	82.80	76.85	78.63	73.61	84.69	57.57	87.31	62.82	83.36
CSI	44.53	92.65	79.08	76.27	75.58	83.78	61.61	86.47	65.20	84.79
ProxyAnchor	87.21	82.43	70.10	79.84	37.19	91.68	65.64	84.99	65.04	84.74
SSD+	31.19	94.19	77.74	79.90	79.39	85.18	66.63	86.18	63.74	86.36
KNN+	39.23	92.78	80.74	77.58	48.99	89.30	57.15	88.35	56.53	87.00
ASH	52.96	90.19	72.62	76.38	75.18	76.52	56.17	86.75	64.23	82.46
CIDER	23.09	95.16	79.63	73.43	16.16	96.33	43.87	90.42	40.69	88.84
CONJ	46.19	90.44	80.81	75.83	60.45	85.90	62.13	83.77	62.40	83.99
DRL	20.15	94.07	76.64	77.55	16.97	94.63	31.97	92.09	36.43	89.59
Medix	0.48	99.87	24.52	93.42	0.73	99.84	8.99	97.92	8.68	97.76
(Ours)	± 0.05	± 0.02	± 1.76	± 0.36	± 0.08	± 0.03	± 0.72	± 0.24	± 0.65	± 0.16

this expanded comparison, including against contrastive-learning baselines, with an average FPR95 of 8.68% and an average AUROC of 97.76%. This result indicates that median filtering provides benefits beyond those obtained by the evaluated representation-learning baselines.

A.4 Evaluation on Large-Scale, Complex Unseen OOD Settings

We next investigate Medix under a large-scale unseen OOD setting. This setting is more challenging for two reasons: (1) it uses a large-scale wild OOD dataset ($\mathbb{P}_{\text{out}}$), increasing both data complexity and computational demand, and (2) the test-time OOD distribution ($\mathbb{P}_{\text{out}}^{\text{test}}$) is intentionally chosen to differ from the wild OOD data used during training, i.e., $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$. We use CIFAR-100 as labeled InD data, the 300K Random Images dataset Hendrycks et al. (2019) as unlabeled wild OOD data, and SVHN as the test OOD data. This setup separates the wild OOD source from the test OOD source, since 300K Random Images differs substantially from SVHN. We compare Medix with baselines that also leverage wild data during training. Table 5 shows that Medix obtains lower FPR95 (41.29 ± 1.2) and higher AUROC (87.25 ± 0.9) than the listed baselines in this unseen-OOD evaluation.

Table 5: OOD detection performance on large-scale unseen OOD data.

Method	FPR ↓	AUROC ↑
OE	68.80 ± 2.8	82.89 ± 1.1
Energy (w/ OE)	69.81 ± 2.4	85.59 ± 1.0
WOODS	69.41 ± 2.7	86.76 ± 0.8
Medix (ours)	41.29 ± 1.2	87.25 ± 0.9

A.5 Evaluating the Impact of Pseudo-Label Quality on OOD Filtering and Robustness

We also examine whether pseudo-label confidence for $\hat{y}_{\tilde{x}_i}$ affects OOD filtering and whether a simple pre-filtering step that removes low-confidence pseudo-labels improves robustness. We conduct this experiment with CIFAR-10 as the InD dataset and LSUN-Resize as the OOD dataset. Specifically, we compute softmax probabilities for each pseudo-labeled sample and discard samples whose confidence falls below 0.6. This filtering removes 15.98% of the samples from the training set.

Performance is nearly unchanged with this additional filtering. Both variants yield FPR95 of 0.01%, while AUROC changes from 99.98% without filtering to 99.99% with filtering. In this setting, low-confidence pseudo-label removal therefore provides little additional benefit, suggesting that the Medix filtering procedure is not highly sensitive to this simple confidence-based refinement.

A.6 Evaluating Computation and Memory Efficiency in Medix Filtering

We next profile the computational and memory cost of the Medix filtering phase. The profiling experiments use an NVIDIA A100-SXM4-80GB GPU and approximately 15,000 unlabeled samples

for each InD–OOD pair. CIFAR-10 and CIFAR-100 serve as InD datasets, and LSUN-Resize serves as the OOD dataset. Table 6 shows modest GPU memory usage and tractable wall-clock times for the evaluated sample sizes. These measurements indicate that the filtering stage is feasible in the reported setting, while larger wild sets would scale with the cost of repeated median computations over the wild gradients.

Table 6: Profiling results for Medix filtering on NVIDIA A100 80GB GPU.

InD–OOD	Wall Clock Time (s)	Peak GPU Memory (MB)	Current GPU Memory (MB)
CIFAR10 – LSUN-Resize	4497.17	99.46	31.74
CIFAR100 – LSUN-Resize	5478.36	99.56	31.79

666

667 A.7 Comparison with Semi-Supervised Open-Set Recognition Methods

Medix differs from recent semi-supervised open-set recognition methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) in objective and mechanism. Methods such as OpenML Saito et al. (2021) address outliers in semi-supervised learning (SSL) while training a standard classifier. In contrast, our setting uses unlabeled wild data to extract candidate OOD samples and train a dedicated OOD detector. Thus, while SSL methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) aim to improve classification despite outliers, Medix targets OOD detection when labeled OOD data is unavailable.

The methods also treat outliers differently. Semi-supervised open-set recognition typically suppresses or downweights OOD samples as noise so that SSL classification improves. Medix instead attempts to identify such samples and then uses them as negative evidence for detector training.

The technical mechanisms are likewise distinct. OpenMatch relies on consistency regularization, enforcing agreement across stochastic transformations of the same input through constraints on classifier outputs. Medix relies on median-based gradient filtering: it uses the robustness of the element-wise median to identify samples whose removal aligns the wild-gradient statistic with the InD reference, and its analysis bounds filtering errors through contamination, concentration, and separation terms.

684 B Broader Impacts and Limitations

OOD detection is a core requirement for reliable machine learning deployment because models may encounter inputs outside the distribution represented by their labeled training data. Medix contributes to this objective by providing a median-centric mechanism for extracting candidate outliers from unlabeled wild data and using them to train an OOD detector. The framework is relevant to safety-sensitive domains in which rejection of unfamiliar inputs is important, while the present study evaluates it in image-classification benchmarks rather than in deployed domain-specific systems. Future work could explore integration with generative models for outlier synthesis or adaptation to dynamic environments where OOD distributions evolve over time.

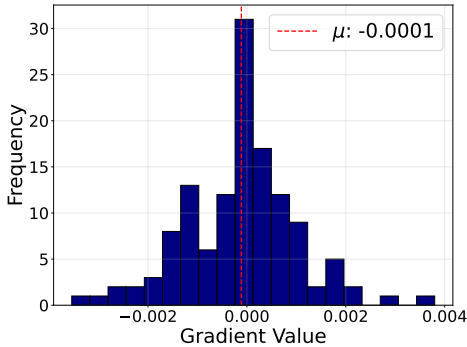
A limitation of Medix is its reliance on an unlabeled wild mixture that contains OOD samples informative enough for detector training. If the wild set contains very few OOD samples, contains OOD samples unrelated to the test-time OOD distribution, or violates the gradient-separation conditions used in the analysis, filtering can become more difficult. The main theorems make these assumptions explicit through contamination, concentration, and separation terms, and Remark 4.3 discusses empirical evidence for the gradient-tail assumption. Theorem C.3 further relaxes the sub-Gaussian condition at the cost of weaker rates. These qualifications define the scope of the current guarantees and point to future work on broader wild-data regimes and more heterogeneous OOD mixtures.

701 C Proof of Theorem

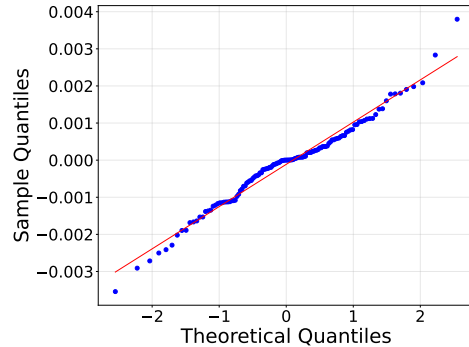
702 For clarity and consistency across the theoretical and empirical sections, Table 7 summarizes the
 703 notation used in the analysis. It includes gradient-related quantities, sample sizes, filtering terms,
 704 and parameters that appear in the concentration bounds. Unless stated otherwise, boldface denotes
 705 vectors, $\mathcal{X}$ denotes the input space, and gradients are computed with respect to a fixed pretrained
 706 model f_ϕ .

Table 7: Notation summary. We use bold symbols for vectors and $\mathcal{X}$ to denote the input space.

Symbol	Meaning
$\mathcal{X}$	Input space (e.g., images in $[0, 1]^d$)
f_ϕ	Model parameterized by ϕ
$\ell(f_\phi(x))$	Loss function evaluated at input x
$\nabla \ell(f_\phi(x))$	Gradient of loss at point x
$\mathbf{g}(x)$	Shorthand for $\nabla \ell(f_\phi(x))$, gradient vector
$\bar{\mathbf{g}}_{\text{in}}$	Empirical mean of gradients over InD points
m	Total number of unlabeled samples in $\mathcal{S}_{\text{wild}}$
$m_{\text{in}} = (1 - \pi)m$	Number of InD samples
$m_{\text{out}} = \pi m$	Number of OOD samples
$\pi \in (0, 1)$	OOD contamination ratio
$\mathcal{S}_{\text{wild}}$	Unlabeled mixture of InD and OOD data
$\mathcal{S}_{\text{in}}^*$	Inliers selected by the median filter
$\mathcal{S}_{\text{out}}^*$	OOD points selected by the median filter
$\mathcal{B}_\varepsilon(\bar{\mathbf{g}}_{\text{in}})$	Ball of radius ε centered at $\bar{\mathbf{g}}_{\text{in}}$
m_ε	Number of InD points inside $\mathcal{B}_\varepsilon$
m_ε^*	Number of selected inliers inside $\mathcal{B}_\varepsilon$
m_{sep}^*	Number of well-separated OODs in selected set
ERR_{in}	Fraction of InD points misclassified as OOD
ERR_{out}	Fraction of OOD points misclassified as InD
σ^2	Sub-Gaussian proxy variance of each gradient coordinate
ε	Threshold for gradient deviation, $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$
$\delta \in (0, 1)$	Confidence parameter for concentration bounds
d	Dimensionality of gradient vectors



(a) InD gradient histogram.



(b) InD gradient Q-Q plot.

Figure 4: Illustration of InD sample gradients in the examined CIFAR-100 setting. (left) Histogram of gradient values showing concentration around the mean with light tails, consistent with sub-Gaussian behavior. (right) Q-Q plot comparing empirical quantiles of InD gradients against a theoretical Gaussian distribution, showing approximate alignment with the Gaussian reference.

707 C.1 Bound on the Inlier Misclassification Rate

708 **Theorem C.1** (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are*
 709 *i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, and*
 710 *fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification*
 711 *rate of the element-wise median (EWM) filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

712 *Proof.* The proof has two steps. In **Step 1**, we establish a high-probability concentration bound for
 713 InD gradients around their mean $\bar{\nabla}_{\text{in}}$ and call an InD sample a *good inlier* when its gradient lies
 714 inside this high-probability region. In **Step 2**, a swapping argument for the EWM objective shows
 715 that removing too many good inliers would contradict optimality of the subset objective. Combining
 716 these steps yields the stated upper bound on ERR_{in} .

717 **Step 1: Concentration of InD Gradients.** Let $\nabla\ell(f_\phi(\mathbf{x})) \in \mathbb{R}^d$ be the gradient (with respect to ϕ)
 718 of a loss evaluated at an InD sample $\mathbf{x}$. Suppose each coordinate of this gradient is sub-Gaussian with
 719 variance proxy σ^2 , centered at the same mean vector $\bar{\nabla}_{\text{in}}$. That is, for each coordinate $j \in \{1, \dots, d\}$,

$$\mathbb{P}\left(|\nabla\ell(f_\phi(\mathbf{x}))_j - \bar{\nabla}_{\text{in},j}| \geq \epsilon\right) \leq 2 \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (6)$$

720 Define

$$\mathbf{v} := \nabla\ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}} \in \mathbb{R}^d.$$

721 It is straightforward to show that

$$\|\mathbf{v}\|_2 \leq \sqrt{d} \|\mathbf{v}\|_\infty,$$

722 so if $\|\mathbf{v}\|_2 > \epsilon\sqrt{d}$ then there must exist at least one coordinate j with $|v_j| > \epsilon$.

723 Hence, combining (6) across $j = 1, \dots, d$ via a union bound yields

$$\mathbb{P}\left(\|\nabla\ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon\sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (7)$$

724 Let $m_{\text{in}} = (1 - \pi)m$ denote the total number of InD points in the set $\mathcal{S}_{\text{wild}}$. We define an InD point
 725 $\mathbf{x}$ to be *bad* if

$$\|\nabla\ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon\sqrt{d}. \quad (8)$$

726 Our goal is to show that, with high probability, only a small fraction of these m_{in} inliers can be bad.

727 From our sub-Gaussian assumption and applying the union bound across the d coordinates, we define
 728 a constant $p \in [0, 1]$ as follows:

$$p := \mathbb{P}\left(\|\nabla\ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon\sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (9)$$

729 Here, $\mathbf{x}$ is a generic InD sample, and we assume that the samples are i.i.d. across the m_{in} inliers.

730 Let X be the random variable denoting the number of bad inliers among the m_{in} samples. For each
 731 inlier $\mathbf{x}_i$, define the indicator variable

$$I_i := \begin{cases} 1, & \text{if } \|\nabla\ell(f_\phi(\mathbf{x}_i)) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon\sqrt{d}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

732 Then, X can be expressed as

$$X = \sum_{i=1}^{m_{\text{in}}} I_i. \quad (11)$$

733 By the linearity of expectation, we have

$$\mathbb{E}[X] = \sum_{i=1}^{m_{\text{in}}} \mathbb{E}[I_i] \quad (12)$$

$$= \sum_{i=1}^{m_{\text{in}}} \mathbb{P}(\mathbf{x}_i \text{ is bad}) \quad (13)$$

$$\stackrel{\text{from (9)}}{\leq} m_{\text{in}} \cdot p \quad (14)$$

$$\leq m_{\text{in}} \cdot 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (15)$$

734 Observing that X is a sum of m_{in} Bernoulli(p) random variables, each bounded between 0 and 1, we
735 can apply Hoeffding's inequality to obtain, for any $t > 0$,

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (16)$$

736 Setting

$$t := \sqrt{\frac{m_{\text{in}} \log\left(\frac{1}{\delta}\right)}{2}}, \quad (17)$$

737 we obtain

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (18)$$

738 With probability at least $1 - \delta$, we therefore have

$$X \leq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (19)$$

739 Dividing both sides by m_{in} yields an upper bound on the fraction of bad inliers:

$$\frac{X}{m_{\text{in}}} \leq \frac{\mathbb{E}[X]}{m_{\text{in}}} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (20)$$

$$\stackrel{\text{from (12)}}{\leq} 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (21)$$

$$= \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}. \quad (22)$$

740 Let us denote this fraction by η . By setting $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$ Thus, with probability at least
741 $1 - \delta$, no more than an $O\left(d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{m_{\text{in}}}}\right)$ fraction of inliers have gradients lying
742 outside the radius $\epsilon\sqrt{d}$ around $\bar{\nabla}_{\text{in}}$. These are precisely the *bad inliers*, while the remainder are
743 called *good inliers*.

744 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
745 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (23)$$

746 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
747 thereby contradicting the optimality of $\mathcal{S}^*$.

748 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 749 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (24)$$

750 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

751 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (25)$$

752 Define the swap size as

$$m'_\varepsilon := \min \{m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (26)$$

753 This quantity counts how many good inliers (those inside B_ε) are currently excluded from $\mathcal{S}^*$ and
 754 available for swapping in.

755 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
 756 and inserting m'_ε good inliers from B_ε in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

757 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (27)$$

758 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (28)$$

759 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (29)$$

760 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (30)$$

761 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 762 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (31)$$

763 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (32)$$

764 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (33)$$

765 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (34)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (35)$$

Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (36)$$

so the points lying inside B_ε form a strict majority in $\hat{\mathcal{S}}$.

Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all gradients in $\mathcal{S}^*$ and $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ the same in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (37)$$

which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (38)$$

For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or increase it, so the median distance from μ_j does not increase and possibly decreases.

Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (39)$$

with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (40)$$

This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes the proof.

$$\Rightarrow \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (41)$$

780

□

781 C.2 Bound on the Outlier Retention Rate

782 We now establish a non-asymptotic upper bound on the fraction of OOD points that remain in
783 the subset returned by the MEDIX filter. This result mirrors the inlier misclassification bound
784 (Appendix C.1), but instead of relying on inlier concentration, it exploits a separation assumption
785 between the OOD and InD gradient means.

786 **Theorem C.2** (Outlier Retention Bound under Vector Separation). *Assume that for every OOD point*
787 *$x \sim P_{\text{out}}$, the gradient $g(x) := \nabla \ell(f_{\varphi_{\text{in}}}(x)) \in \mathbb{R}^d$ satisfies:*

788 **(I) Sub-Gaussianity.** *Each coordinate $g_j(x)$ is sub-Gaussian with variance proxy σ_{out}^2 .*

789 **(II) Vector Mean Separation.** *The mean OOD gradient vector μ_{out} satisfies*

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d}, \quad \text{for some } \Delta > 0.$$

790 Fix any tolerance $\varepsilon \in (0, \Delta)$ and confidence parameter $\delta \in (0, 1)$. Define:

$$p_{\text{out}}(\varepsilon) := 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right), \quad \eta_{\text{out}} := p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}.$$

791 Then with probability at least $1 - \delta$, the outlier retention rate satisfies:

$$\text{ERR}_{\text{out}} := \frac{|\mathcal{S}_{\text{in}}^* \cap \text{OOD}|}{m_{\text{out}}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}. \quad (42)$$

792 **Proof. Step 1: Concentration of OOD Gradients.**

793 By assumption, each coordinate of $g(x)$ is sub-Gaussian, and the mean vector is separated from $\bar{\nabla}_{\text{in}}$
 794 by at least $\Delta\sqrt{d}$. Therefore, for any OOD point x , the event

$$\|g(x) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \quad (43)$$

795 has probability at most $p_{\text{out}}(\varepsilon)$.

796 **Counting the “ambiguous” OOD points.** For every OOD example x_i define the indicator

$$J_i := \mathbf{1}\left\{\|g(x_i) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}, \quad i = 1, \dots, m_{\text{out}}.$$

797 Each $J_i \sim \text{Bernoulli}(p_{\text{out}}(\varepsilon))$ and the J_i ’s are independent. Let

$$Y := \sum_{i=1}^{m_{\text{out}}} J_i, \quad \mu := \mathbb{E}[Y] = m_{\text{out}} p_{\text{out}}(\varepsilon).$$

798 Because the indicators J_i take values in $\{0, 1\}$, Hoeffding’s inequality states that for any $s > 0$

$$\Pr\{Y - \mu \geq s\} \leq \exp\left(-\frac{2s^2}{m_{\text{out}}}\right).$$

799 Choose $s = \sqrt{\frac{m_{\text{out}} \log(1/\delta)}{2}}$; then (C.2) equals δ , and with probability at least $1 - \delta$

$$Y \leq m_{\text{out}} \left(p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}\right) = m_{\text{out}} \eta_{\text{out}}.$$

800 Define the high-probability event

$$\mathcal{E}_1 := \{Y \leq m_{\text{out}} \eta_{\text{out}}\}, \quad (35)$$

801 which holds with probability at least $1 - \delta$.

802 **Step 2: Contradiction via a Swapping Argument** Suppose, toward contradiction, that

$$\text{ERR}_{\text{out}} > \eta_{\text{out}} + \frac{1 - \pi}{2\pi}. \quad (44)$$

803 Let $m_{\text{out}}^* := |\mathcal{S}_{\text{in}}^* \cap \text{OOD}|$, and define $m_\varepsilon^* := \#\{\text{ambiguous OODs in } \mathcal{S}_{\text{in}}^*\}$. Then, under $\mathcal{E}_1$,

$$m_{\text{sep}}^* := m_{\text{out}}^* - m_\varepsilon^* > \frac{1 - \pi}{2\pi} m_{\text{out}} = \frac{1}{2} m_{\text{in}}. \quad (45)$$

804 Thus, more than half the points in $\mathcal{S}_{\text{in}}^*$ are separated OODs whose gradients lie outside the ball

$$B_\varepsilon := \left\{g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}. \quad (46)$$

805 By the definition of the coordinate-wise median, this implies that in every coordinate, the majority of
 806 entries in $G_{\mathcal{S}_{\text{in}}^*}^{(j)}$ lie outside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, hence:

$$\|\text{EWM}(G_{S_{\text{in}}^*}) - \bar{\nabla}_{\text{in}}\|_2^2 \geq \varepsilon^2 d. \quad (47)$$

Now consider a new subset $\hat{S}$ formed by swapping the m_{sep}^* separated OODs in S_{in}^* with an equal number of InD points whose gradients lie inside B_ε . Such inliers exist because fewer than $m_{\text{out}}\eta_{\text{out}}$ are ambiguous, and the total number of InD points is m_{in} .

Then at least half the entries in every coordinate of $G_{\hat{S}}^{(j)}$ lie inside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, implying:

$$\|\text{EWM}(G_{\hat{S}}) - \bar{\nabla}_{\text{in}}\|_2^2 \leq \varepsilon^2 d. \quad (48)$$

Combining with (47) yields a contradiction:

$$\|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\hat{S}})\|_2^2 < \|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{S_{\text{in}}^*})\|_2^2. \quad (49)$$

This contradicts the optimality of S_{in}^* , so the assumed violation cannot hold. With probability at least $1 - \delta$, the outlier retention rate is therefore bounded by

$$\text{ERR}_{\text{out}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}, \quad (50)$$

□

C.3 Theorem C.3 without Sub-Gaussianity

Theorem C.3 (Inlier Misclassification Bound without Sub-Gaussianity). *Assume that the gradients of InD points in S_{wild} are i.i.d., and that each coordinate has variance σ^2 and finite fourth moment bounded by μ_4 , i.e.,*

$$\mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^2 = \sigma^2, \quad \mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^4 \leq \mu_4.$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the number of InD samples in S_{wild} , and fix any tolerance $\varepsilon > \sigma$ and confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the element-wise median (EWM) filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq \underbrace{2 \left(\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \right)}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1 - \pi)}}_{\text{Contamination term}}.$$

Differences from the sub-Gaussian bound. The sub-Gaussian bound features a decay term

$$2 \exp\left(-\frac{\varepsilon^2 d}{2\sigma_{\text{sub}}^2}\right)$$

that decreases exponentially with dimension d . In contrast, the finite-fourth-moment bound has the decay term

$$\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2}$$

which decays as $O(1/d)$ and is therefore looser. The sub-Gaussian bound requires light tails, whereas the finite-fourth-moment version allows heavier-tailed gradient coordinates. Both bounds share the finite-sample term

$$\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}},$$

that shrinks as m_{in} increases, independent of the tail assumptions.

830 *Proof.* The proof follows the structure of the original Theorem 4.1, consisting of two main steps: (1)
 831 establishing a concentration bound on the fraction of “bad” InD samples (those far from ∇_{in}) without
 832 relying on sub-Gaussianity, and (2) using a swapping argument to bound the inlier misclassification
 833 rate. The key difference is the replacement of the sub-Gaussian tail bound with a moment-based
 834 bound derived from the finite fourth moment assumption.

835 **Step 1: Concentration of InD Gradients** Define the gradient deviation for an InD sample x as:

$$v = \nabla \ell(f_\phi(x)) - \nabla_{\text{in}} \in \mathbb{R}^d, \quad (51)$$

836 where $\nabla_{\text{in}} = \frac{1}{n} \sum_{(x_i, y_i) \in S_{\text{in}}} \nabla \ell(f_\phi(x_i))$ is the mean gradient over the labeled InD training set S_{in} ,
 837 and we assume ∇_{in} is the population mean. An InD sample is classified as “bad” if:

$$\|v\|_2 > \epsilon \sqrt{d}, \quad (52)$$

838 and “good” otherwise. Our goal is to bound the fraction of bad inliers among the m_{in} InD samples in
 839 S_{wild} .

840 Since $\|v\|_2^2 = \sum_{j=1}^d v_j^2$, we analyze the ℓ_2 norm via the sum of squared coordinates. Assume that
 841 the coordinates $v_j = \nabla \ell(f_\phi(x))_j - \nabla_{\text{in},j}$ are independent across $j = 1, \dots, d$, each with:

$$\mathbb{E}[v_j] = 0 \quad (\text{assuming } \nabla_{\text{in},j} = \mathbb{E}[\nabla \ell(f_\phi(x))_j]), \quad (53)$$

$$\mathbb{E}[v_j^2] = \sigma^2, \quad \mathbb{E}[v_j^4] \leq \mu_4. \quad (54)$$

842 Thus:

$$\mathbb{E}[\|v\|_2^2] = \mathbb{E}\left[\sum_{j=1}^d v_j^2\right] = d\sigma^2, \quad (55)$$

$$\text{Var}(\|v\|_2^2) = \text{Var}\left(\sum_{j=1}^d v_j^2\right) = \sum_{j=1}^d \text{Var}(v_j^2). \quad (56)$$

843 Compute the variance of v_j^2 :

$$\text{Var}(v_j^2) = \mathbb{E}[v_j^4] - (\mathbb{E}[v_j^2])^2 \leq \mu_4 - \sigma^4, \quad (57)$$

844 so:

$$\text{Var}(\|v\|_2^2) \leq d(\mu_4 - \sigma^4). \quad (58)$$

845 Using Chebyshev’s inequality for $\|v\|_2^2$:

$$\mathbb{P}(|\|v\|_2^2 - d\sigma^2| \geq t) \leq \frac{d(\mu_4 - \sigma^4)}{t^2}. \quad (59)$$

846 We need $\mathbb{P}(\|v\|_2 > \epsilon \sqrt{d})$, which corresponds to $\|v\|_2^2 > \epsilon^2 d$. Set the threshold:

$$\|v\|_2^2 > \epsilon^2 d \implies \|v\|_2^2 - d\sigma^2 > \epsilon^2 d - d\sigma^2 = d(\epsilon^2 - \sigma^2). \quad (60)$$

847 Thus:

$$\mathbb{P}(\|v\|_2^2 > \epsilon^2 d) = \mathbb{P}(\|v\|_2^2 - d\sigma^2 > d(\epsilon^2 - \sigma^2)) \leq \frac{d(\mu_4 - \sigma^4)}{[d(\epsilon^2 - \sigma^2)]^2} = \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}, \quad (61)$$

848 provided $\epsilon > \sigma$ so that $\epsilon^2 - \sigma^2 > 0$. Define:

$$p = \mathbb{P}\left(\|\nabla \ell(f_\phi(x)) - \nabla_{\text{in}}\|_2 > \epsilon\sqrt{d}\right) \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (62)$$

849 Let X be the number of bad inliers among the m_{in} InD samples in S_{wild} , where each InD sample is
850 bad with probability at most p . Then:

$$\mathbb{E}[X] \leq m_{\text{in}} p \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (63)$$

851 Since $X = \sum_{i=1}^{m_{\text{in}}} I_i$, where $I_i = 1$ if the i -th InD sample is bad and 0 otherwise, and I_i are i.i.d.
852 Bernoulli(p) variables bounded in $[0, 1]$, apply Hoeffding's inequality:

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (64)$$

853 Set $t = \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}$, so:

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (65)$$

854 With probability at least $1 - \delta$:

$$X \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (66)$$

855 The fraction of bad inliers is:

$$\eta = \frac{X}{m_{\text{in}}} \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}. \quad (67)$$

856 This η represents the upper bound on the fraction of InD samples whose gradients deviate from ∇_{in}
857 by more than $\epsilon\sqrt{d}$, relying only on finite fourth moments and coordinate independence.

858 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
859 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1 - \pi)}, \quad (68)$$

860 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 861 thereby contradicting the optimality of $\mathcal{S}^*$.

862 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 863 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (69)$$

864 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

865 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (70)$$

866 Define the swap size as

$$m'_\varepsilon := \min \{m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (71)$$

867 This quantity counts how many good inliers (those inside B_ε) are currently excluded from $\mathcal{S}^*$ and
 868 available for swapping in.

869 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
 870 and inserting m'_ε good inliers from B_ε in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

871 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (72)$$

872 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (73)$$

873 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (74)$$

874 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (75)$$

875 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 876 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (76)$$

877 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (77)$$

878 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (78)$$

879 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (79)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (80)$$

880 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (81)$$

881 so the points lying inside B_ε form a strict majority in $\widehat{\mathcal{S}}$.

882 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
883 gradients in $\mathcal{S}^*$ and $\widehat{X}^{(j)} = \{\widehat{x}_1^{(j)}, \dots, \widehat{x}_m^{(j)}\}$ the same in $\widehat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

884 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
885 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
886 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\widehat{X}^{(j^*)}) \in I_{j^*}, \quad (82)$$

887 which implies

$$|\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (83)$$

888 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
889 increase it, so the median distance from μ_j does not increase and possibly decreases.

890 Thus, component-wise:

$$|\text{EWM}_j(G_{\widehat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (84)$$

891 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\widehat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (85)$$

892 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
893 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (86)$$

894

□